

Title:**Procedure**

The development of the Alumni Information System follows the approach of the Systems Development Life Cycle (SDLC) using the Agile methodology to ensure flexibility, collaboration, and iterative progress toward a user-centered solution.

Managing software development can be challenging due to changing requirements, technology upgrades, and cross-functional collaboration. The software development life cycle (SDLC) methodology provides a systematic management framework with specific deliverables at each stage of the software development process. As a result, all stakeholders agree in advance on software development goals and requirements and also have a plan to achieve these goals. (Mehmet,2023).

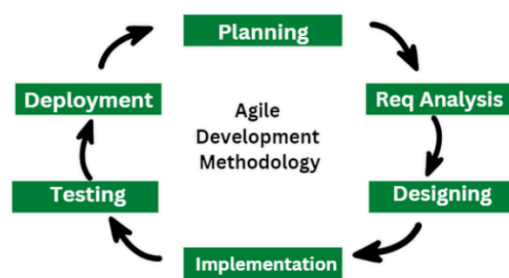


Figure 2. Agile Development

Planning. The development process begins with careful planning, where the primary goals, scope, and objectives of the Alumni

Information System are established. This involves identifying the system's key functionalities, such as alumni data management, usability features, and reporting capabilities. A product backlog is created to prioritize tasks and determine the iterative sprints needed for the development. Communication with Alumni Relations Directors ensures that the system's objectives align with user needs.

Requirement Analysis. The next phase involves analyzing the requirements gathered during planning. Detailed specifications for the system are outlined, focusing on features like database design, user interface, and usability enhancements. Functional and non-functional requirements are identified, ensuring the system meets both user expectations and technical standards. Feedback from Alumni Relations Directors is continuously sought to validate the requirements and refine the development roadmap.

Designing. In the design phase, the conceptual framework of the Alumni Information System is developed. Wireframes and prototypes are created to visualize the system's layout and user interface. The design focuses on simplicity and accessibility to accommodate users across the six campuses. Flowcharts and diagrams are used to illustrate system workflows, ensuring that all features align with the defined requirements. Alumni

Relations Directors review the prototypes to provide input for necessary adjustments before development begins.

Implementation. During the implementation stage, the system is developed incrementally in sprints. Core modules, such as the alumni database and user registration features, are built first, followed by additional functionalities like reporting tools and analytics. Each sprint delivers a working increment of the system, which is immediately available for feedback and testing. The researcher ensures code quality and functionality throughout the implementation phase.

Testing. Testing is an integral part of the Agile cycle, conducted after each sprint and at the end of the implementation phase. The system is subjected to unit testing, integration testing, and user acceptance testing to identify and resolve issues. Selected alumni and Alumni Relations Director test the system's usability, providing valuable feedback to refine features. Bugs and inconsistencies are addressed promptly to maintain the system's quality.

Deployment. Once the system passes all testing stages, it is deployed for live use across the six BISU campuses. During

deployment, stakeholders and users interact with the system to ensure it functions as intended.

Iterative Improvement. The Agile methodology emphasizes continuous improvement, even after deployment. Feedback collected during the live deployment phase is used to refine the system further. Periodic updates and enhancements are made based on user needs, ensuring the Alumni Information System evolves to remain relevant and effective for BISU's alumni management goals.

This iterative and flexible approach ensures that the system is developed in a user-centered manner, addressing the unique needs of BISU's alumni community while maintaining high quality and efficiency.